Geometry of Superposition in a Toy Model and SAE Recovery

Homework 7 — Research in LLMs, HSE x Central University

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Abstract

I study how a toy ReLU autoencoder (F features compressed into d dimensions) learns superposition, and how well a vanilla Sparse Autoencoder recovers the ground-truth features. Three findings stand out. First, at d=2 the model discovers clean polytope structures (digons, triangles, pentagons) that transition discretely as F increases. Second, SAE reconstruction quality (EV > 0.97 for most configurations, dropping only under aggressive sparsity) is a poor proxy for feature recovery: high EV coexists with weighted MMCS as low as 0.80. Third, rare features are systematically lost by the SAE, with a sharp frequency threshold separating recovered from unrecovered features.

1 Setup

The toy model and SAE follow the assignment specification exactly. One implementation note: the EV formula includes a square root, $\mathrm{EV} = 1 - \sqrt{\mathrm{Var}(h - \hat{h})/\mathrm{Var}(h)}$, which is stricter than the standard R^2.

Weighted MMCS. Vanilla MMCS is misleading when some features have near-zero norm. If the model assigns $\|W_i\| \approx 0$ to a rare feature (effectively not representing it), the cosine similarity with any dictionary element is undefined or arbitrary — yet it still contributes equally to the average. I introduce weighted MMCS: each feature's max-cosine is weighted by $\|W_i\|/\max_j \|W_j\|$. Features the model does not represent contribute proportionally less. In practice, weighted MMCS and vanilla MMCS agree when F/d is small (all features represented), but diverge at high superposition where many features have negligible norm.

Experimental details. Default configuration: F=50, d=5, α=1.0, 10k toy model training steps, 20k SAE steps, batch size 1024, Adam optimizer, F_{SAE}=100, ℓ_0=0.1, 10% L1 warmup. All sweeps use 3 seeds (42, 43, 44) and report mean ± std.

2 Part 1: Geometry of Feature Embeddings

2.1 Polytope Structures at d=2

Hypothesis: When d=2, the model should discover discrete geometric arrangements to pack $F > 2$ features into a 2D space, forming structures analogous to solutions of the Thomson problem (placing repulsive charges on a circle).

I trained models with $F \in \{4, 5, 6, 8, 10\}$ at d=2, α=1.0.

The model reliably discovers structured arrangements (Figure 1). At F=4, all three seeds produce two antipodal pairs (digons) — the optimal packing of 4 directions in 2D. At F=5, the model finds a pentagon or a digon-plus-triangle arrangement depending on the seed. At F=6, three digons or two triangles appear. All three seeds produced structured arrangements at these small F values.

A key observation: point sizes in Figure 1 reveal that the model does not represent all features equally. At F=8 and F=10, rare features (small circles, yellow) are pushed to near-zero norm while frequent features (large circles, purple) maintain large norms.

SAE recovery in the d=2 regime is near-perfect: weighted MMCS exceeds 0.998 for all F values (Table 1), — clean geometry makes the SAE's job easy.

Table 1: d=2 polytope sweep (mean ± std, 3 seeds).

F	EV	wMMCS	Dead%	FRR
4	.987 ± .002	**.998** ± .003	38%	1.00
5	.987 ± .003	**1.00** ± .001	22%	1.00
6	.986 ± .004	**1.00** ± .000	32%	1.00
8	.988 ± .001	**1.00** ± .000	22%	0.96
10	.989 ± .001	**.999** ± .001	23%	1.00

2.2 Power-Law Exponent Controls Representation Allocation

Hypothesis: Higher α steepens the frequency distribution, concentrating $\|W_i\|$ on the most frequent features and leaving rare features unrepresented.

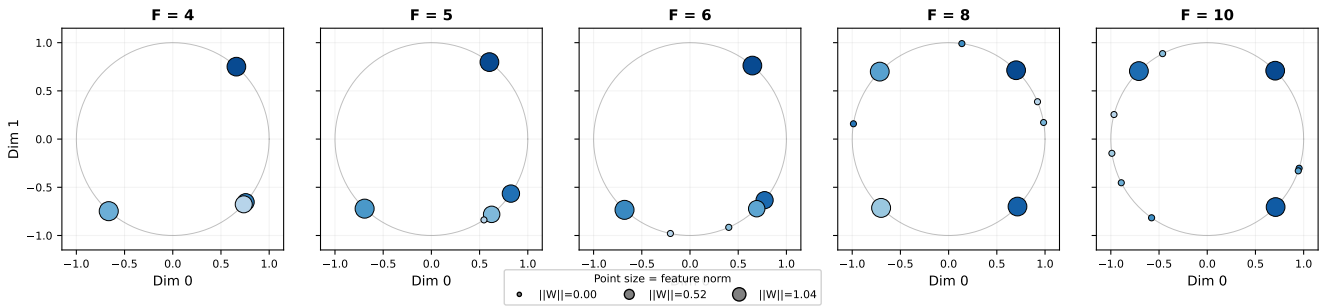

Figure 1: Feature embeddings on the unit circle for d=2, α=1.0, seed 42. Point size is proportional to $\|W_i\|$; color encodes feature index. The model discovers antipodal pairs (digons) at F=4, and progressively denser angular packings as F increases.

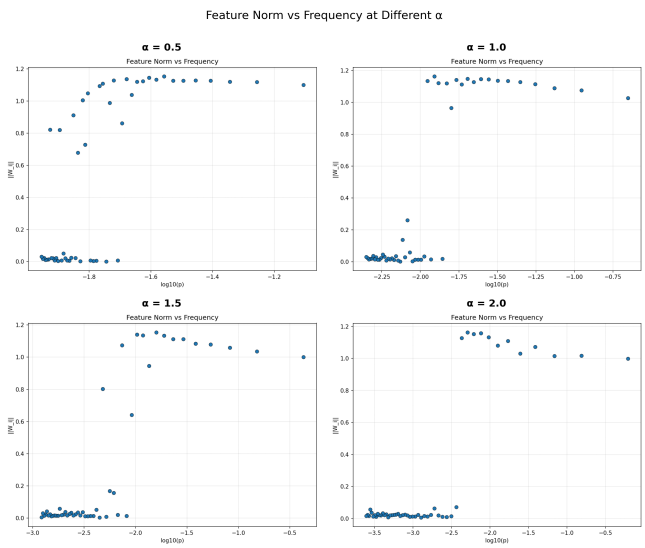

Figure 2: $\|W_i\|$ vs $\log_{10}(p_i)$ for $\alpha \in \{0.5, 1.0, 1.5, 2.0\}$, F=50, d=5.

Figure 2 shows this directly. At α=0.5 (flat distribution), feature norms spread across a wide range with no clear threshold. At α=2.0, there is a sharp step: the top ∼10 features have $\|W_i\| \approx 0.7$–0.9, and the rest collapse to $\|W_i\| < 0.1$.

Higher α also improves weighted MMCS: 0.911 ± 0.009 at α=0.5 versus 0.969 ± 0.007 at α=2.0 (Table 2). Higher α improves wMMCS despite sacrificing more features: the survivors have high $\|W_i\|$ and well-separated directions, which the SAE finds easily.

Note that $\mathbb{E}[\|f\|_0] = \sum_i p_i = 1.0$ for all α by construction (the normalization constant C ensures this). The discriminating quantity is the entropy $H(p) = -\sum p_i \log p_i$, which drops from 3.77 at α=0.5 to 1.52 at α=2.0 (Table 2). Lower entropy means more concentrated activations: at α=2.0, the top 5 features account for 90% of the probability mass. Recovery improves because the distribution is more peaked, not because it is sparser.

Table 2: α sweep, F=50, d=5 (mean ± std, 3 seeds). $H(p)$: entropy of feature probability distribution.

α	$H(p)$	EV	wMMCS	FRR
0.5	3.77	.987	$.911 \pm .009$	0.53
1.0	3.20	.989	$.937 \pm .007$	0.68
1.5	2.30	.990	$.955 \pm .006$	0.73
2.0	1.52	.991	$\mathbf{.969} \pm .007$	0.80

2.3 Cosine Similarity Characterizes Superposition Regime

As F/d increases, the pairwise cosine similarity distribution of W columns transitions from structured to random-like (Figure 3).

Figure 3 shows the transition. At F=10 (F/d=2), the distribution is wide with visible structure — some feature pairs are near-orthogonal, others near-parallel. At F=50 (F/d=10), the distribution narrows and centers near zero. At F=100 (F/d=20), it becomes approximately Gaussian with standard deviation ∼0.25.

The feature dimensionality metric $D_i = \|W_i\|^2 / \sum_j (\hat{W}_j \cdot W_i)^2$ tells the same story (Figure 4). At F=50, d=5, most features have D_i below 0.1, far below the dedicated (D=1), digon (D=0.5), and triangle (D=2/3) reference lines. The model operates firmly in deep superposition.

2.4 Training Dynamics

The assignment asks how training steps affect geometry. I trained the toy model (F=50, d=5, α=1.0) for 10k steps and examined W at checkpoints 500, 1k, 2k, 5k, and 10k (Figure 5).

The loss converges by approximately 2k steps (Figure 5, left), dropping from 10^{-2} to $\sim 2.2 \times 10^{-3}$ and changing negligibly thereafter. The geometric structure, however, continues evolving: the model starts by representing 38 features at step 500, then progressively discards rare features, stabilizing at 18 represented features

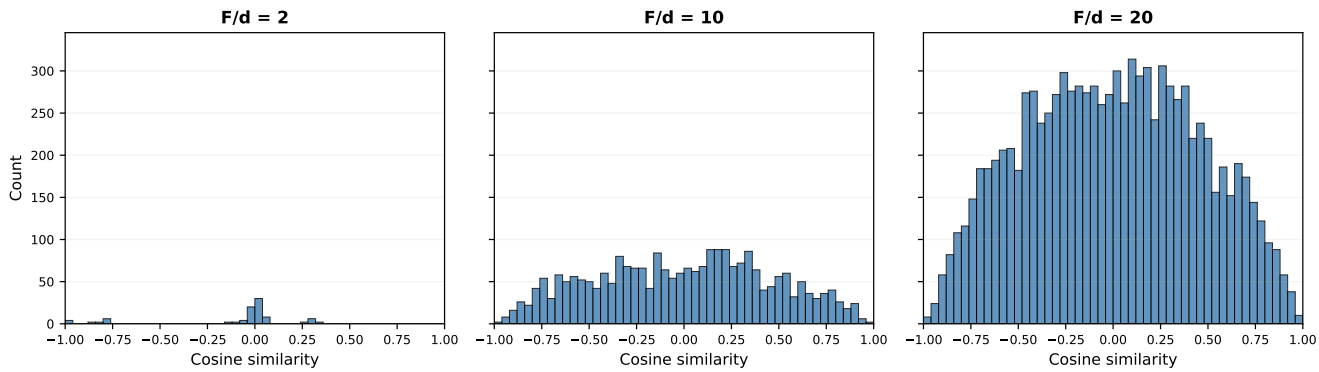

Figure 3: Off-diagonal cosine similarity distributions at F/d = 2, 10, and 20 (d=5, α=1.0). Shared y-axis scale across panels.

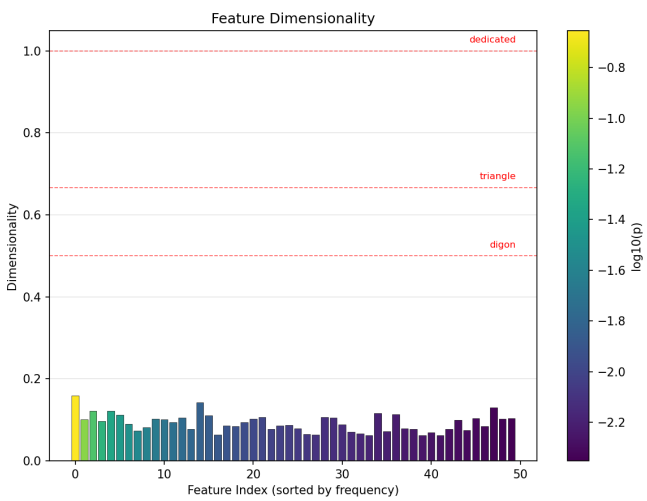

Figure 4: Per-feature dimensionality for F=50, d=5, α=1.0. Horizontal lines mark dedicated (D=1), digon (D=0.5), and triangle (D=2/3) regimes.

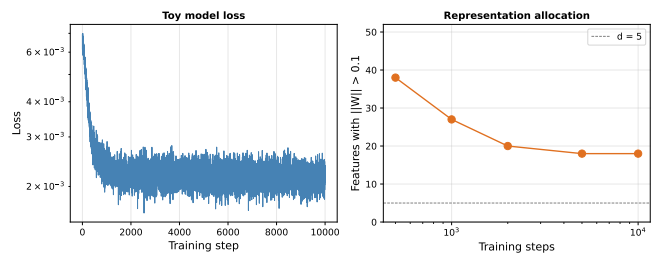

Figure 5: Left: toy model loss over 10k steps. Right: number of features with $\|W_i\| > 0.1$ at each checkpoint.

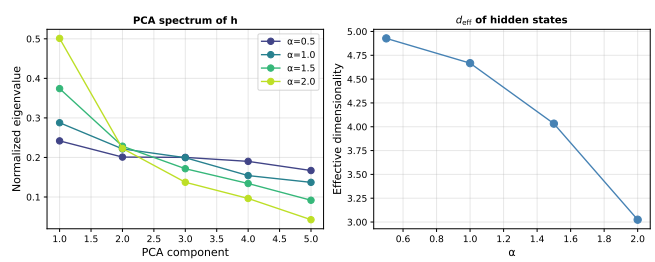

Figure 6: Left: PCA spectrum of h for each α (F=50, d=5). Right: effective dimensionality d_{eff} (participation ratio) vs α.

by step 5k. So loss converges first (∼2k steps), but the model keeps pruning features for another ∼3k steps after that. The final count of 18 represented features, well above d=5, confirms that the model operates in the superposition regime even after convergence.

2.5 Hidden State Structure

The hidden states $h = W^T f$ inherit structure from both the feature distribution and the learned embedding. Figure 6 shows the PCA spectrum of h for different α values.

At α=0.5, variance is distributed nearly uniformly across all d=5 components ($d_{\text{eff}} = 4.93$). At α=2.0, the first component captures 50% of variance and the last under 5% ($d_{\text{eff}} = 3.02$). This reflects the concentration: at high α, only a few features contribute meaningfully to h, so the hidden state occupies a lower-dimensional subspace. Lower d_{eff} means the SAE searches a smaller space, explaining the better recovery at high α (Table 2).

3 Part 2: SAE Reconstruction Quality

3.1 The Sparsity–Fidelity Tradeoff

Hypothesis: There exists a Pareto frontier between reconstruction quality (EV) and sparsity (L_0), controlled by the ℓ_0 coefficient.

Figure 7 and Table 3 confirm this. EV peaks at 0.995 near ℓ_0=0.01 and degrades at higher sparsity, reaching 0.922 ± 0.002 at ℓ_0=1.0. The slight improvement from ℓ_0=0.001 to 0.01 suggests mild L1 regularization benefits reconstruction. The L_0 (mean active latents) drops correspondingly from 80.9 to 11.5.

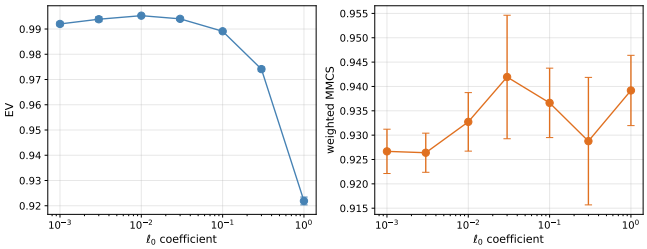

Figure 7: Left: EV vs ℓ_0. Right: weighted MMCS vs ℓ_0. EV spans a 7.3 pp range while wMMCS varies by only 1.6 pp (F=50, d=5, α=1.0).

Table 3: Pareto frontier: ℓ_0 sweep at F=50, d=5 (3 seeds).

ℓ_0	EV	wMMCS	Dead%	L_0	FRR
0.001	.992	.927	0.0	80.9	0.79
0.003	.994	.926	0.0	50.8	0.75
0.01	.995	.933	0.0	34.1	0.75
0.03	.994	.942	0.3	30.0	0.74
0.1	.989	.937	3.7	20.9	0.68
0.3	.974	.929	8.3	15.9	0.61
1.0	.922	.939	15.0	11.5	0.67

Weighted MMCS is stable across the full sweep: 0.926 to 0.942, a 1.6 percentage point range versus 7.3 pp for EV. The SAE learns the same decoder directions regardless of ℓ_0.

3.2 Dictionary Size Scaling

Larger dictionaries should help — more capacity to decompose superposed representations.

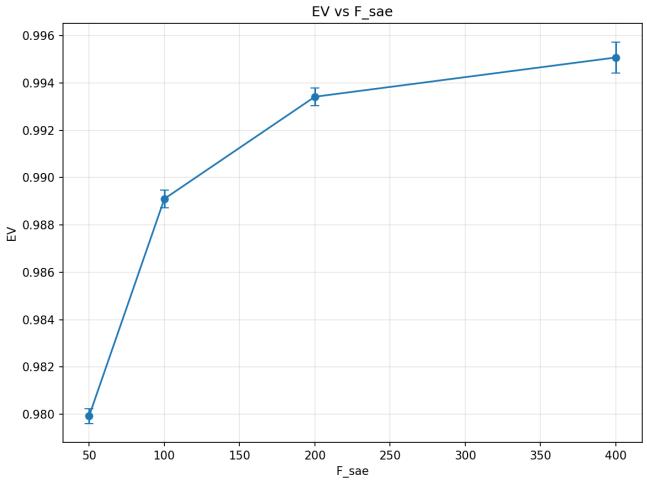

Figure 8: EV vs F_{SAE} (F=50, d=5, α=1.0, ℓ_0=0.1).

EV improves from 0.980±0.000 at F_{SAE}=50 to 0.995± 0.001 at F_{SAE}=400 (Figure 8). More importantly, feature recovery rate (fraction with cosine > 0.9) jumps from 0.44 at F_{SAE}=50 to 1.00 at F_{SAE}=400 (Table 4). MMCS improves correspondingly from 0.883 to 0.963.

Table 4: Dictionary size sweep (F=50, d=5, 3 seeds).

F_{SAE}	EV	wMMCS	MMCS	FRR
50	.980 ± .000	.903 ± .006	.883 ± .007	0.44
100	.989 ± .000	.937 ± .007	.921 ± .003	0.68
200	.993 ± .000	.954 ± .002	.945 ± .002	0.89
400	.995 ± .001	**.963** ± .009	.963 ± .005	1.00

At F_{SAE}=400 (8× overcomplete), every ground-truth feature is recovered above the 0.9 cosine threshold. The price is a high dead latent fraction (23.9%) — most of the 400 dictionary elements go unused.

3.3 EV Is Not Discriminative

Across all experiments, EV exceeds 0.97 for every configuration except the most aggressive sparsity penalty (ℓ_0=1.0: EV = 0.922) and d=10 with high ℓ_0 (Table 5). So EV cannot distinguish an SAE that recovers everything from one that misses half the features.

The assignment asks whether properties of the learned W matrix predict SAE quality. Figure 9 plots EV against two summary statistics of W across all 66 (config, seed) runs: mean absolute pairwise cosine similarity (measuring feature interference) and effective rank (measuring how many dimensions W effectively uses).

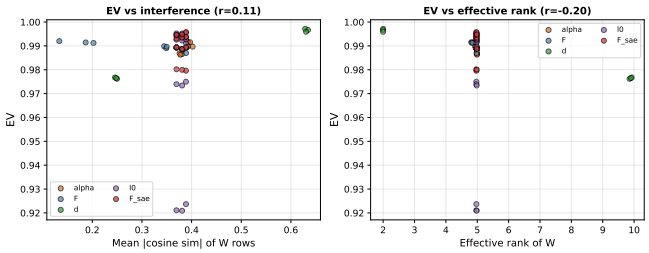

Figure 9: EV vs properties of W across all sweep configurations. Each point is one (config, seed) run, colored by sweep.

Neither metric correlates meaningfully with EV (r = 0.11 for interference, r = −0.20 for effective rank). EV is primarily controlled by the SAE's own hyperparameters (ℓ_0, F_{SAE}) rather than by the toy model's geometry. Again: EV does not tell you whether the SAE found the right features.

The d=10 regime is notable (Table 5): at the default ℓ_0=0.1, weighted MMCS is only 0.803±0.012 with a feature recovery rate of 18.7%. Increasing ℓ_0 to 1.0 improves weighted MMCS to 0.881±0.025 and recovery to 52.7%, at the cost of EV dropping from 0.977 to 0.832. At d=10 you need higher ℓ_0 — the default 0.1 is not enough.

Synthesis. EV above 0.97 is achievable across nearly all configurations tested. The primary controls are the

Table 5: d=10 operating point search (F=50, 3 seeds).

ℓ_0	EV	wMMCS	MMCS	FRR
0.1 (default)	.977	.803	.802	0.19
0.3	.942	.817	.816	0.25
0.5	.908	.830	.829	0.33
1.0	.832	**.881**	.877	0.53

SAE's own parameters: ℓ_0 (which trades reconstruction for sparsity) and F_{SAE} (which provides capacity). Model parameters (α, F/d) have minimal effect on EV but dramatically affect feature recovery (Section 4). Properties of W (interference, effective rank) do not predict EV either. So EV does not reflect feature recovery quality.

4 Part 3: Feature Recovery

4.1 Rare Features Are Systematically Lost

Hypothesis: The SAE preferentially recovers frequent features (high p_i, high $\|W_i\|$) and systematically fails on rare features.

Figure 10 makes this clear. At F=10 (F/d=2), every feature achieves cosine similarity > 0.95 with its matched dictionary element. At F=100 (F/d=20), the bar plot shows a clear bifurcation: the first ∼15 features (highest frequency) are recovered well, while the remaining ∼85 features have cosine similarities scattered between 0.6 and 0.95, with many below the 0.9 threshold.

The frequency recovery scatter (Figure 11, left) shows the bifurcation: at default dictionary size (F_{SAE}=100), roughly 30 of 50 features cluster near the diagonal while the rest have SAE latents that never fire. At 8× overcomplete (Figure 11, right), nearly all features are recovered.

Feature recovery rate drops sharply with superposition depth: 1.00 at F/d=2, 0.68 at F/d=10, and 0.62 at F/d=20 (Figure 12, left). At F/d=20, the SAE recovers only 62% of features.

4.2 ℓ_0 Controls the Recovery Threshold

While Section 3.1 showed that decoder directions are stable across ℓ_0, the feature recovery rate varies substantially: 0.79 at ℓ_0=0.001 down to 0.61 at ℓ_0=0.3.

Low ℓ_0 produces many active latents (L_0 = 80.9), meaning the SAE distributes each hidden state across many dictionary elements. This improves reconstruction but does not align dictionary elements with individual features. Medium ℓ_0 (∼0.01–0.03) forces the SAE to use fewer, more interpretable directions, improving per-feature alignment at minimal cost to EV.

4.3 Superposition Depth Limits Recovery

The dominant constraint is F/d. Weighted MMCS degrades from 0.989 at F/d=2 to 0.923 at F/d=20 (Figure 12, left). The degradation is smooth and consistent across 3 seeds.

The d sweep reveals an asymmetry (Table 6): d=2 achieves near-perfect recovery (wMMCS = 1.000) while d=10 struggles (wMMCS = 0.803), even though both have F/d=25 and F/d=5 respectively. The d=10 regime has more room for superposition to spread features across dimensions in non-axis-aligned ways, making SAE recovery harder.

Table 6: Hidden dimension sweep, F=50, α=1.0 (3 seeds).

d	F/d	EV	wMMCS	MMCS	FRR
2	25	$.997 \pm .001$	$\mathbf{1.00} \pm .000$	.9997	1.00
5	10	$.989 \pm .000$	$.937 \pm .007$	.921	0.68
10	5	$.977 \pm .000$	$.803 \pm .012$	.802	0.19

4.4 EV and Feature Recovery Are Decoupled

Consider two configurations: at d=2, EV = 0.997 with weighted MMCS = 1.000 and FRR = 1.00. At d=10, EV = 0.977 with weighted MMCS = 0.803 and FRR = 0.19.

The EV difference is 2 percentage points. The feature recovery difference is 81 percentage points. An SAE can explain 97.7% of variance in the hidden states while recovering fewer than 1 in 5 ground-truth features above the 0.9 cosine threshold.

Similarly, in the ℓ_0 sweep: EV spans 0.922–0.995 (7.3 pp range) while weighted MMCS spans 0.926–0.942 (1.6 pp range) and FRR spans 0.61–0.79 (18 pp range). The metrics probe different aspects of SAE quality.

EV alone would have hidden the poor recovery in most of my experiments.

5 Discussion

The 2D geometry is clean, and superposition scales smoothly with F/d and α. The SAE recovers features reliably when superposition is mild ($F/d \leq 2$) but degrades systematically at higher superposition ratios.

The main result is the EV–MMCS decoupling. EV can be near-perfect (> 0.97) while feature recovery is poor (wMMCS < 0.85). In my experiments, EV alone would have been misleading for almost every configuration.

The d=10 anomaly is not fully explained by these experiments. The fact that d=2 (higher F/d=25) achieves perfect recovery while d=10 (F/d=5) struggles suggests that the difficulty of SAE recovery depends not just on

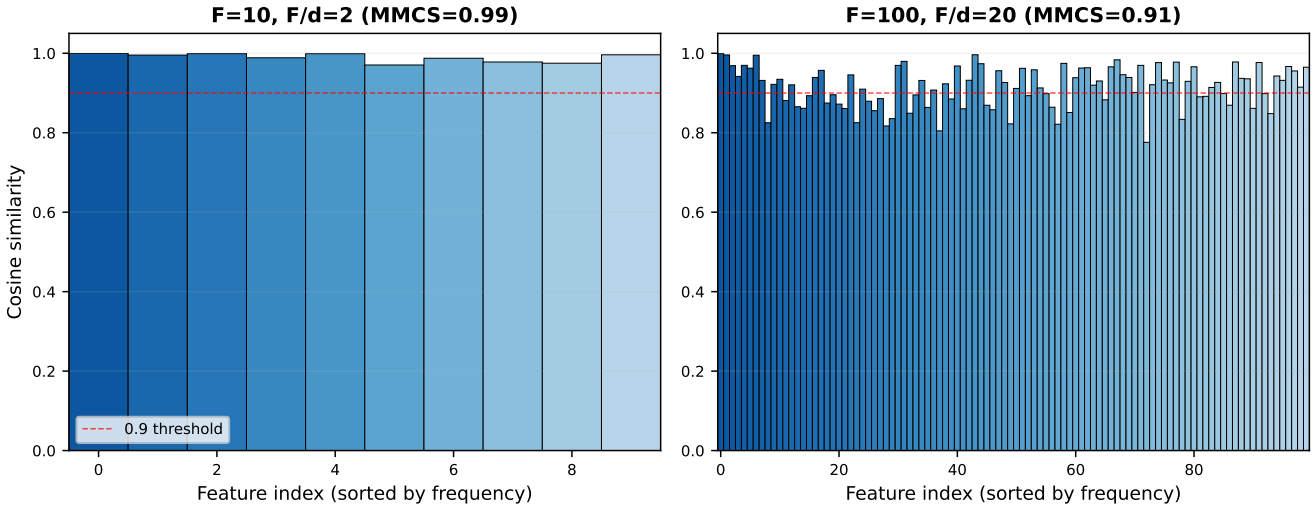

Figure 10: Per-feature cosine similarity with matched SAE latent. Left: F=10 (F/d=2, near-perfect). Right: F=100 (F/d=20, selective recovery). Dashed line: 0.9 recovery threshold. Color encodes $\log_{10}(p_i)$.

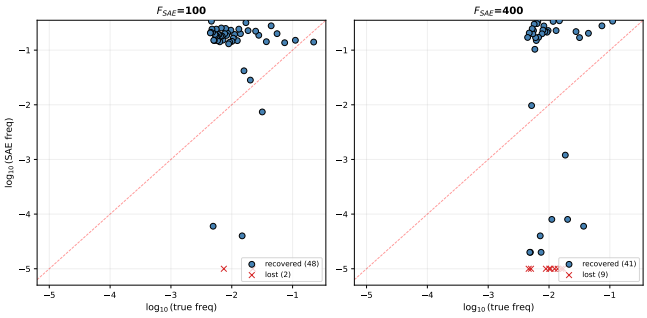

Figure 11: Frequency recovery scatter. Left: F_{SAE}=100 (default). Right: F_{SAE}=400 (8× overcomplete). At default dictionary size, rare features (high index, yellow) cluster at the bottom — their matched SAE latents never fire. At 8× overcomplete, most features are recovered near the diagonal. Note: "recovered" here means the matched SAE latent has nonzero activation rate, not the cosine > 0.9 threshold used for FRR in the tables.

the superposition ratio but on the absolute dimensionality. In 2D, features must lie on a circle — a 1D manifold — severely constraining the geometry and making it easy for the SAE to find. In 10D, the space of possible feature arrangements is vastly larger. This is a curse-of-dimensionality effect.

Limitations. This study uses uniform importance ($I_i = 1$), a vanilla SAE architecture (no TopK, no Gated SAE), and relatively short training (10k–20k steps). All experiments run on a single toy model class with power-law features. All of this is on a toy model with independent Bernoulli features — real transformer activations have correlated features and non-uniform magnitude distributions, so the specific numbers here should not be

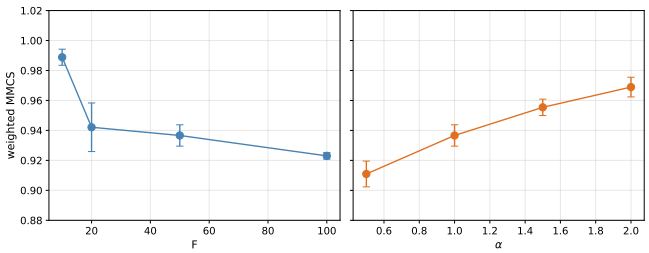

Figure 12: Weighted MMCS vs F (left) and vs α (right). Mean ± std over 3 seeds. Y-axis zoomed to 0.88–1.02.

taken literally.

Possible extensions. If I had more time, I would try inverse-frequency importance weighting in the toy model loss — the current uniform $I_i = 1$ systematically disadvantages rare features, and fixing that might be the cheapest way to improve tail recovery.